EXPERIMENT-2

06/08/26

STUDY OF DIFFERENT TYPES OF NETWORK CABLES

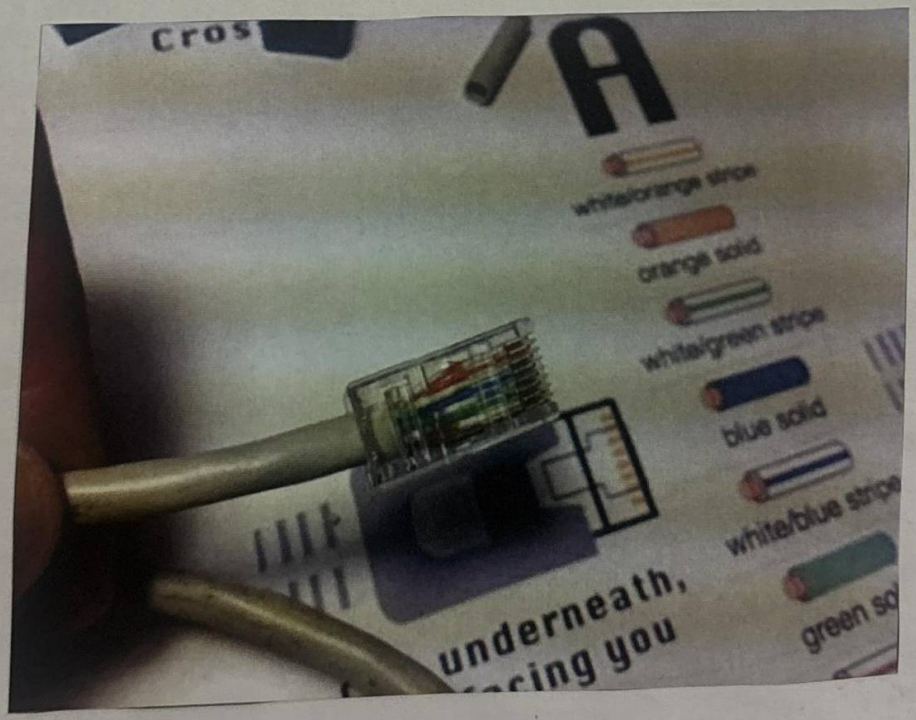
AIM:

To understand the different types of Network cables:

The different types of cables used in networking are:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Cable (STP)
3. Coaxial cable
4. Fibre Optic Cable

CABLE TYPE	CATEGORY	MAXIMUM DATA TRANSMISSION	ADVANTAGES/DISADVANTAGES	APPLICATION/USE
UTP	Category 3	10 bps	Advantages: → Cheaper in cost → Easy to install as they have a smaller overall diameter. Disadvantages: → More prone to (EMI)	10-Base-T Ethernet Fast Ethernet, Gigabit Ethernet Fast Ethernet, Gigabit Ethernet.
	Category 5	Upto 100 Mbps		
	Category 5E	1 Gbps		
STP	Category 6, 6a	10 Gbps	Advantages: → Shielded → Faster than UTP. → Less susceptible to noise & interference.	Gigabit Ethernet, 10GB Ethernet (55 m) widely used in data centres.
SSTP	Category 7	10 Gbps	Disadvantages: → Expensive → Greater installation effort	Gigabit Ethernet, 10G Ethernet (100 m)


Cros
A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
underneath,
facing you

Coaxial cable	RG-6, RG-59, RG-11	10-100 Mbps	→ High bandwidth → Immune to interference → Low loss bandwidth → Versatile Disadvantages: → Limited distance → Low loss bandwidth → Size is bulky	Speed of signal is 500m Television network. High speed internet connections.
Fiber Optics Cable	Single mode Multi mode	100 Gbps	Advantages: → High speed, bandwidth, security, long distance. Disadvantages: → Expensive → Requires skilled installers	Maximum distance of fibre cable is around 100 meters.

PROCEDURE:

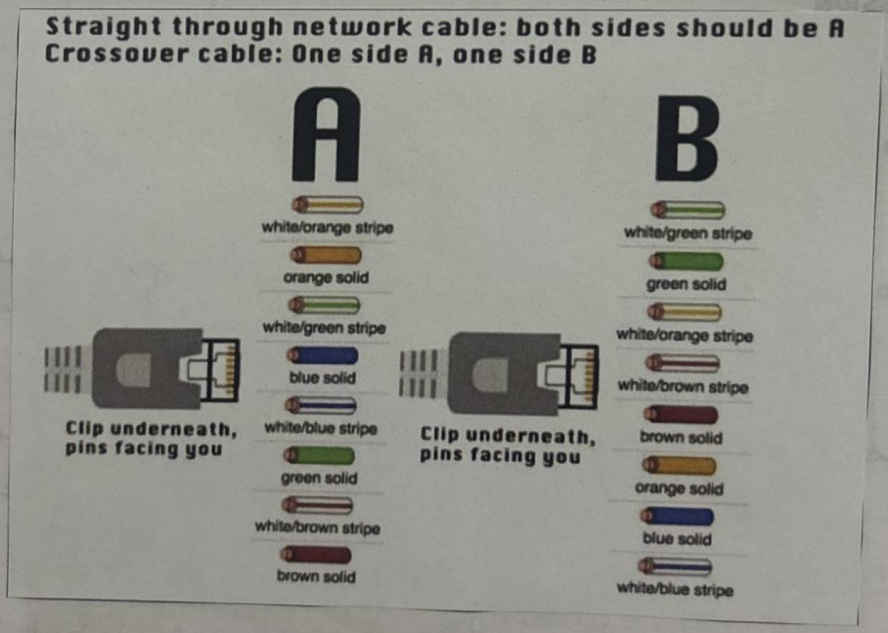
1.) Prepare the materials: Ethernet cable, crimping tools, two RJ45 connectors & optional connector boots/shields.

2.) Insert the cable boot: Slide the protective shield onto the cable before attaching the RJ45 connector.

3.) Untwist the wires: Separate the 4 twisted pairs & carefully straighten the eight individual wires.

4.) Arrange the wires: Hold the eight wires together in the correct order & cut their ends evenly.

5.) Insert into the RJ45 connector: Hold the RJ45 plug with the gold contacts during upward you & carefully insert all 8 wires while maintaining their order.


Straight through network cable: both sides should be A
Crossover cable: One side A, one side B
A
white/orange stripe
orange solid
white/green stripe
blue solid
white/blue stripe
green solid
white/brown stripe
brown solid
Clip underneath, pins facing you
B
white/green stripe
green solid
white/orange stripe
white/brown stripe
brown solid
orange solid
blue solid
white/blue stripe
Clip underneath, pins facing you

6.) Crimp the connector: Place the RJ45 connector into the crimping tool & press firmly to secure the wires & contacts.

7.) Test the cable: Connect the cable b/w 2 compatible network devices & check whether the devices establish a network connection.

RESULT:

Thus, the different types of network cables have been studied and an ethernet straight-through/cross-over cable was successfully prepared & tested.